2023

(Session : 2022-25)

(Paper ID : 11201)

Time : 3 hours Full Marks : 80

Note: *Candidates are required to give their answers in their own words as far as practicable.*
The questions are of equal value.
Answer any five questions.

1. Define the following with an example :

   a. Singleton Set
   b. Proper Subset
   c. Complement of Set
   d. Venn-Diagram

2. Given universal set $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, $A = \{2, 4, 6\}$, $B = \{1, 3, 5, 7\}$, $C = \{6, 7\}$, then find :

   a. $A^c \cap B$
   b. $(A \cup B) - C$
   c. $(A \cup C)^c$
   d. $(A \cup U) \cap (B \cup C)$

3. What is function ? Define it. Explain various types of functions with suitable example.

4. If $f(x) = ax^2 + bx + 2$, $f(1) = 3$ and $f(4) = 42$, find b.

5. If $A = \{1, 3, 5\}$, let R be a relation such that XRY : if $y = x + 2$ and S be the relation such that XSY : if $x < y$ then find following:

   a. R o S

b. S o R

c. R o R

d. S o S

With the help of diagram.

6. What is Graph ? Explain sequential representation of graph (directed, undirected, weighted) using adjacency matrix with suitable example.

7. What is Algebraic structure if Discrete Mathematics ? Explain properties of algebraic structure with suitable example.

8. What is Hash Diagram ? Show that set of all divisors of 12 forms a lattice.

9. Let the function of $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^2 + 1$. Find $f^1(-5)$, $f^1(26)$ and $f^1(10, 37)$.

10. Define following:

a. Partition of a Set

b. Semi Group

c. Binary Operation

d. POSET

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